Statistics Interview Questions (with Simple Answers)

1. What is the difference between descriptive and inferential statistics?

Descriptive statistics summarize and describe the main features of a dataset. Inferential statistics use sample data to make conclusions or predictions about a larger population.

2. What are mean, median, and mode?

Mean is the average, median is the middle value, and mode is the most frequent value. Median is preferred when data is skewed.

3. What is variance and how is it different from standard deviation?

Variance measures the average squared deviation from the mean. Standard deviation is the square root of variance, giving spread in original units.

4. What are the characteristics of a normal distribution?

Symmetrical, bell-shaped, mean = median = mode, follows the 68-95-99.7 rule.

5. What is a sampling distribution?

It is the distribution of a statistic (like the mean) based on many samples from the same population.

6. Explain the Central Limit Theorem.

It states that the sampling distribution of the sample mean becomes normal as the sample size increases, even if the original data is not normal.

7. What is the difference between population and sample?

Population is the whole group; sample is a smaller subset used for analysis.

8. What are null and alternative hypotheses?

Null hypothesis (H0) is the default assumption; alternative hypothesis (H1) is what you want to prove.

9. What is a p-value?

It is the probability of observing the data if the null hypothesis is true.

10. What are Type I and Type II errors?

Type I: Rejecting true H0 (false positive). Type II: Not rejecting false H0 (false negative).

11. What is statistical power?

The probability of correctly rejecting a false null hypothesis (1 - beta).

12. Difference between covariance and correlation?

Covariance shows direction of relationship; correlation shows both direction and strength (standardized).

13. What does a correlation coefficient of 0, 0.5, 1 mean?

0: no correlation, 0.5: moderate positive correlation, 1: perfect positive correlation.

14. What are assumptions of linear regression?

Linearity, independence, homoscedasticity, normality of residuals, no multicollinearity.

15. What is multicollinearity?

When independent variables in regression are highly correlated with each other.

16. What is a Q-Q plot?

A plot to check if data follows a normal distribution by comparing quantiles.

17. What are skewness and kurtosis?

Skewness shows asymmetry; kurtosis shows heaviness of tails.

18. When to use box plots, histograms, or scatter plots?

Box plots: show spread and outliers. Histograms: show distribution. Scatter plots: show relationships.

19. What is a confidence interval?

A range of values that likely contains the population parameter.

20. What is a confidence level?

The probability that the confidence interval contains the true parameter (e.g., 95%).

21. Difference between random and purposive sampling?

Random: everyone has equal chance. Purposive: selected based on characteristics.

22. Why is sampling important?

Because studying the entire population is often impractical or impossible.

23. Role of statistics in machine learning?

Used for understanding data, feature selection, evaluation, and making predictions.

24. How to test if a dataset is normally distributed?

Use Q-Q plots, histograms, or statistical tests like Shapiro-Wilk.

25. Real-world example of using statistics?

Example: Survey analysis, predicting heart disease, or A/B testing in marketing.